

Midterm 1 Part b

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I. QUESTION (9 POINTS)

Give an example dataset of 12 instances as a two class problem, where the first class has 8 and the second one 4 instances. There will be only one feature (continuous) in the dataset.

- Draw a bootstrap sample from it.
- Draw a 3 fold cross-validation (without stratification) sample. Show all three folds.
- Draw a 2 fold cross-validation (with stratification) sample. Show all two folds.

II. QUESTION (12 POINTS)

Assume that you have a 2 dimensional 2 class dataset. For the following algorithms, give an example estimated covariance matrix S (or matrices S_i depending on the algorithm). Also plot the covariance matrices.

- Quadratic Discriminant Analysis
- Linear Discriminant Analysis
- Naive Bayes
- K-means

III. QUESTION (10 POINTS)

The following table shows the first and second exam grades obtained for students in a database course.

First	Second
5	11
4	7
3	5

Derive a linear equation for the prediction of a student's second exam grade based on the student's first grade in the course. (Hint: Do linear regression where x is the midterm and y is the final grade. The function will be $y = ax + b$)

IV. QUESTION (10 POINTS)

Given the following confusion matrix:

	Predicted Class	P	N
True Class			
P		50	40
N		10	80

Calculate accuracy, error, fp-rate, precision, and recall.